

The electricity generated by PV panels is direct current (DC), which needs to be converted into alternating current (AC) using an inverter.

Panels are mounted on the rooftop using special mounting structures. If solar power is required when sunlight is not enough for the panels to generate electricity (such as at night), battery backup is required.

Types of solar PV systems. Grid-tied systems are a combination of solar PV and grid. As these systems depend on the grid-supplied reference voltage for their operation, grid-tied systems do not work when the grid is down.

Grid-interactive systems work in conjunction with either battery backup or diesel generator to support loads even during power failure.

Off-grid systems operate independent of the grid. These systems sync with diesel generators and/or batteries. Battery-based systems were uncommon in the past due to their high costs. However, the situation is changing now, given the improved

Warranty

Various components of rooftop solar installations come with varying periods of warranty:

PV panels. Industry-standard warranty is 10 years on the product and 25 years on performance.

Other systems. Inverters, mounting structures, cables, junction boxes, etc typically come with a 1-5-year product warranty, which is extendible subject to conditions.

economics of battery technology.

Types of mount. Mounting systems for rooftop installations are available for both inclined and flat roofs.

A variety of mounting systems are available, the most common being railed, rail-less and shared rail. All these systems require some type of penetration or anchoring onto the roof, whether attached to rafters or directly to the decking. The standard residential systems use rails attached to the roof to support rows of solar panels. Each panel, usually positioned vertically/

portrait-style, attaches to two rails with clamps. The rails secure to the roof by a type of bolt or screw, with flashing installed around/over the hole for a watertight seal.

Business economics

Solar business models include:

1. Service-based operation. Types of services being solar rooftop power plant installation and maintenance

2. Operational model. Just-in-time sourcing of components and products as per the project requirement

3. Financial model. A low-cost model where investment is done for mainly fixed and variable cost heads. Raw material or tools are purchased after order confirmation by the customer, utilising the advance from the customer.

Fixed-cost items include:

1. Company registration
2. Solar installation training
3. Certification
4. Miscellaneous fixed assets (laptop, smartphone, etc as per requirement)



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